

Aditya Ahirrao

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Computer Science & Design undergraduate specializing in full-stack development, artificial intelligence, and machine learning, with hands-on experience building RAG-based LLM systems, RESTful APIs, and scalable full-stack applications.

Education

K. K. Wagh Institute of Engineering Education and Research

B.Tech in Computer Science and Design; CGPA: 8.5

Nashik, India

Aug 2023 – 2027 (Expected)

Services Preparatory Institute

Higher Secondary Certificate Examination

Chhatrapati Sambhajinagar, India

2023

P. S. B. A. English School

Secondary School Examination

Chhatrapati Sambhajinagar, India

2021

Experience

AI & Machine Learning Intern

June 2026 – July 2026

Hindustan Aeronautics Limited (HAL)

Nashik, India

- Engineered a Retrieval-Augmented Generation (RAG) pipeline with semantic search and vector embeddings, enabling natural-language question answering across **10,000+ pages** of aircraft maintenance manuals and accident reports.
- Built the retrieval layer using ChromaDB and Sentence Transformers, backed by a FastAPI service integrated with PostgreSQL for document indexing.
- Applied Docling to structure unstructured text, tables, and technical drawings for ingestion into the pipeline.
- Deployed locally hosted LLMs via Ollama to generate citation-backed responses, and built an analytics dashboard to track usage.
- Continued the project independently after the internship as **AeroLens**, an offline documentation assistant.

Digital Marketing & Content Strategist

Feb 2026 – June 2026

WRS Energy Solutions

Remote

- Applied digital marketing and SEO fundamentals to plan and curate audience-focused content aligned with brand messaging, driving a **24.4% net growth** in targeted Instagram followers over a 4-month period.

UI/UX Designer & Researcher

Nov 2025 – Jan 2026

Runverve | Smart Heal (Sponsored, IIT Madras-incubated startup)

Remote

- Conducted user research and usability testing to design high-fidelity UI/UX prototypes in Figma.
- Integrated third-party RESTful APIs into a React Native massage-therapy app, improving engagement and accessibility.

Projects

Swasthyam | React Native, Node.js, MongoDB, ESP32, Python, REST APIs

Nov 2025 – Present

- Led UI/UX design and frontend development for an AI/IoT healthcare platform built for production-grade scalability.
- Integrated ESP32 sensor data via REST APIs to power real-time monitoring and lipid-profile-based recommendations.

Health Companion | Python, FastAPI, Tkinter, Telegram Bot API

June 2026 – Present

- Independently built** a healthcare management desktop app with secure FastAPI backend services.
- Automated medication reminders for patients using the Telegram Bot API.

Carbon Footprint Platform | HTML, CSS, JavaScript, Python, Git

April 2026 – May 2026

- Developed a sustainability analytics platform for survey-based data collection, emissions calculation, and reporting.
- Piloted across an educational society's 3 colleges (1 engineering, 2 pharmacy); results cross-checked against published benchmarks of 1,500–4,500 tCO₂e/year.

Technical Skills

- Languages:** C, C++, Python, JavaScript, TypeScript, HTML, CSS
- AI/ML:** PyTorch, YOLO (Computer Vision), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Natural Language Processing (NLP)
- Full-Stack & Backend Development:** FastAPI, React, React Native, Node.js, RESTful API Design
- Databases:** PostgreSQL, MongoDB, ChromaDB (Vector Database)
- Tools & Platforms:** Git/GitHub, Figma, Canva, ESP32 (IoT)

Achievements, Leadership & Certifications

- Winner**, Smart India Hackathon 2025 – out of 400 competing teams nationwide (Dec 2025).
- National Qualifier**, IUCEE Student Project Competition (2026).
- Runner-Up**, Hackathon @ IIIT Gwalior, Madhya Pradesh – among 30 national teams (Apr 2026).
- President**, Design Society (Aug 2025 – Present).
- Head**, INNOVERA Hackathon – led a team of 80 (Jan – Mar 2025).
- Chairperson, Branding**, YI KKWIEER (Dec 2025 – Present); **Design Team Member**, TEDx.
- Certifications:** Programming and DSA using Python, Problem Solving through C (NPTEL – **Course Topper, Top 5%**); Digital Marketing and Web Development (MKCL).
- Interests:** Marathon Running, Cycling, Calisthenics, Reading.